SOVEREIGN GRID: 3D PLANETARY & SQUARE ECOSYSTEM

Master Project Specification, Game Design & 3D Model Architecture Plan (Compiled from Full Chat)

1. Executive Vision & Core Game Loop

Sovereign Grid unifies a 3D Human-Scale Solar System, a Free Fire (FF)-style Open Planetary Map (82% Forest + 18% Oceans), and Clash-of-Clans (CoC) Fortified Squares with a gamified mastery and strategy ecosystem.

* Focused 1-Planet Strategy: Perfected deeply on Planet 1 (Gaia) first so all FF horseback travel, globe joystick controls, CoC Squares, Floating Sky Main Halls, and Giant 3D Tracker Rooms work seamlessly before expanding unique infrastructures to the remaining 7 planets.

2. 3D Solar System Layer (Orbit View - Human Solar System Scale)

* Human Solar System Planet Sizes (Earth Landmass Preserved): All 8 planets in Orbit retain their Earth-class continents, specular oceans, cloud decks, and natural moons, while their 3D radii and diameters match the exact Human Solar System hierarchy:
* - Orbit 01 | Gaia (Home) : Mercury-Scale (4,879 km diameter, 3D radius 1.85) - 82% Emerald Forest + 18% Ocean
* - Orbit 02 | Olympus : Venus-Scale (12,104 km diameter, 3D radius 4.55) - Highland Earth Continents & Seas
* - Orbit 03 | Elysium : Earth-Scale (12,742 km diameter, 3D radius 4.80) - Verdant Rainforest Continents
* - Orbit 04 | Atlantis : Mars-Scale (6,779 km diameter, 3D radius 2.55) - Pelagic Island Continents
* - Orbit 05 | Arcadia : Jupiter-Scale (139,820 km diameter, 3D radius 11.20) - Largest Giant Earth World
* - Orbit 06 | Hyperborea : Saturn-Scale (116,460 km diameter, 3D radius 9.45) - 2nd Largest Giant Earth World
* - Orbit 07 | Avalon : Uranus-Scale (50,724 km diameter, 3D radius 6.80) - Ice-Giant Sized Earth World
* - Orbit 08 | Aether : Neptune-Scale (49,244 km diameter, 3D radius 6.55) - Outer Giant Earth World
* Orbit Navigation: Click the Orbit button in the top-right bar at any time to jump to the 3D Solar System, fly in 6DOF Spaceship Cockpit mode, or select any planet to drop back onto its surface.

3. Free Fire (FF)-Style Planetary Surface Map (82% Forest)

* Solid 3D Spherical Earth: Continuous 3D planet terrain with 82% emerald forest landmass coverage, golden coastal beaches, turquoise shallows, and 18% deep sapphire ocean basins.
* GPU-Instanced 3D Forest (tree_forest.glb): Your custom Google Drive tree model is GPU-instanced hundreds of times across the 82% forest continent at 60 FPS using THREE.InstancedMesh.
* Far-Apart Continental Places: Places are distributed across distant regions of the globe (Sovereign Sanctum HQ, Asia Silk Citadel, Americas Sierra Redoubt, Smriti Library, Ares War Camp, Temporal Crucible, Arka Treasury Vault, Pelagic Trade Harbor).
* Horseback Travel & Dual Joystick Controls: Ride a 3D horse from place to place using WASD / Arrow Keys or the on-screen Horse Ride Joystick (with Gallop and Rear/Jump buttons), or toggle to Rotate Globe Mode to spin and inspect the planet from above.
* FF Planet Radar Minimap: Live bottom-left tactical radar displaying your heading, all

planetary places, and one-click horse travel or Square entry.

4. Dummy Level Monuments Visible from Above Earth

* Every place on the planet surface projects a 3D Tiered Holographic Dummy Level Totem (e.g., Lv.14, Lv.17) visible from high above Earth.
* Logging progress in shops, training army units, or placing Clan War trophies dynamically increases the place Dummy Level and visual tier.

5. Clash-of-Clans (CoC) Squares, Floating Sky Main Hall & Sky Stairs

* Fortified CoC Square Layout: Every place on the planet has an enterable CoC Square featuring a checkerboard lawn, perimeter walls, specialized Shops, and a waiting Warhorse at the NW trail gate to ride back to the planet.
* FF Walk View vs. CoC Top View: Walk, sprint, and jump on foot inside every Square like in Free Fire (3rd-person chase camera) or switch to CoC Top View (isometric overview).
* Floating Sky Main Hall (main_hall.glb) Outside the Square Box: Your custom 3D Main Hall model hovers high in the air (Y = 7.2, Z = -24.5) outside the northern gate of every Square Box, supported by a floating sky-island rock spire and glowing anti-gravity levitation rings.
* 24-Step Climbable 3D Sky Staircase: A golden walkway leads from the central square compass (0,0,0) to the northern gate (Z = -12), where a 24-step illuminated 3D Sky Staircase rises through the air to the Floating Sky Main Hall. In FF Walk View, walking North through the gate physically climbs all 24 steps into the sky onto the floating Main Hall terrace.

6. Library Square, Tracker Shops & FF Training Room (Colossal 3D Tracker)

* Library Square Tracker Shops: Entering Smriti Library Square gives access to 4 specialized shops to log details across multiple trackers:
* 1. Mistake Autopsy Shop - Error patterns, silly mistakes, and root-cause fixes.
* 2. Exam & Mock Score Shop - Full-length mock test scores, percentile trends, and accuracy.
* 3. Syllabus Codex Shop - Chapter completion, revision passes, and mastery checkpoints.
* 4. Deep Drill Speed Shop - Timed problem batches and speed-drill logs.
* Log Progress -> FF Training Room Scene: Clicking Log Progress in any shop transitions into a 3D FF Training Room where a 3D player character stands holding a glowing logging device in front of a colossal, towering 3D Holographic Tracker Monolith (trackers look massive in this scene). While the 3D character beams data into the monument, you log your real entries via the console.

7. Army Place & Troop Dispatch System

* Train and expand your planetary armed forces in the Army Camp across 3 unit tiers: Vanguard Infantry, Siege Breakers, and Plasma Lancers.
* Dispatch trained armies to fortify planetary places or launch orbital attacks on mineral-rich asteroids.

8. Money & Mining Square, Asteroid Tracker & Live 3D Asteroid Scene

* Money & Mining Square: Mine CE currency, collect stores, visit resource shops, and train Mining Engineers equipped with heavy extraction tools.
* Interactive Asteroid Mineral Tracker: Tracks near-orbit mineral asteroids (16-Psyche Core Shard, Kepler-Veld Cobalt Belt, Aetherium Monolith, Chronos Sovereign Vein).
* Two-Stage Asteroid Operations:
* - Phase 1 (Army Combat Strike - ~5 Minutes): Dispatched Army fights to clear orbital

defenses.

* - Phase 2 (Engineer Extraction - 5 Hours to 5 Days): Once combat succeeds, Mining Engineers deploy laser drills to extract minerals over 5 hours to 5 days based on level and difficulty.
* Live 3D Asteroid Scene (Go to Asteroid 3D): Visit any asteroid in 3D at any time to watch the live orbital battle or 3D Engineer laser-drilling rigs extracting minerals.

9. Clan War Rewards & Monument Placement

* Win Clan War trophies (Golden Siege Obelisk, Inferno Dragon Brazier, Aether Crystal Pylon, Sovereign Grand Banner) and place or relocate them freely either inside any CoC Square or directly on the Main Planet Surface.

10. Headquarters (Main Hall Square), Trusts & Weekly 3D Council Meeting

* Headquarters Trust Buildings: Houses embassy buildings for the different Trusts from all places across the planet (Library Knowledge Trust, Kshatriya Defense Trust, Dhana Treasury Trust, Tarka Logic Trust).
* Weekly Interactive 3D AI Council Meeting: 5 animated 3D Delegate models gather around a central holotable and speak in turn, presenting only basic public info of the planet and all places (forest coverage, Main Hall levels, army readiness, and asteroid mining status) while never exposing your private logged entries.

11. Custom External 3D Model (.glb) Pipeline & Next Models Checklist

* INTEGRATED: tree_forest.glb -> GPU-instanced across 82% of Planet Gaia forest landmass.
* INTEGRATED: main_hall.glb -> Floating Sky Main Hall outside every Square Box + on Planet Surface with 24-step Sky Stairs.
* READY TO DROP IN NEXT (via Google Drive link):
* - Green & Icy Mountains (.glb) for alpine ridges and polar peaks on the planet surface.
* - Shop & Trust Buildings (.glb) for Library, Money/Mining, and Headquarters Squares.
* - Warhorse & Rider (.glb) for planetary horseback travel and Square gates.
* - Player & Army Troop Units (.glb) for FF walking, Training Room, and Army Camp.
* - Asteroid Drilling Rigs & Clan War Trophies (.glb).